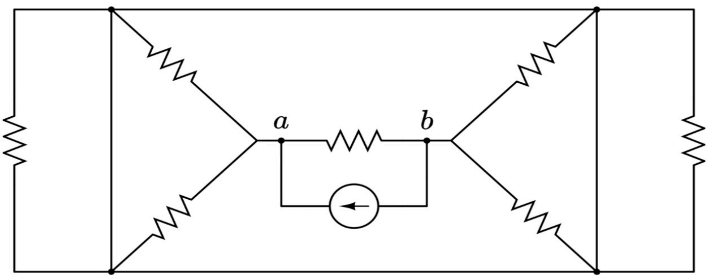
Practice Quiz Two: Equivalent Resistance (Hard)

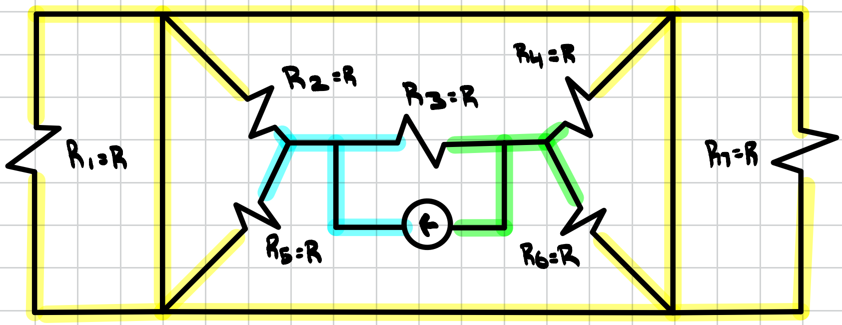
For the circuit shown below, where every resistor has an identical resistance R, find the equivalent resistance R_{ab} seen by the independent current source. Express your answer in terms of R.

Suggested time limit: 15 minutes



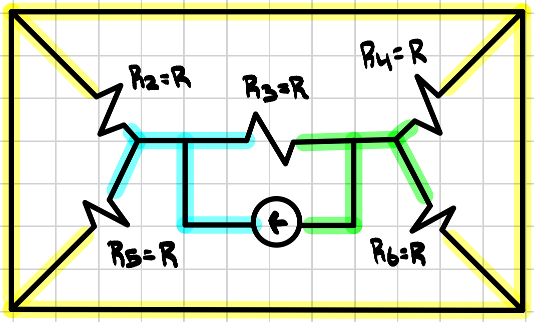
I will use label like R_1 and R_2 so you know which resistor I am refering to but all have a Value of R

Start by labeling nodes which are wires between at least two components



Cyan = Node A
Green = Node B
Yellow = Node C

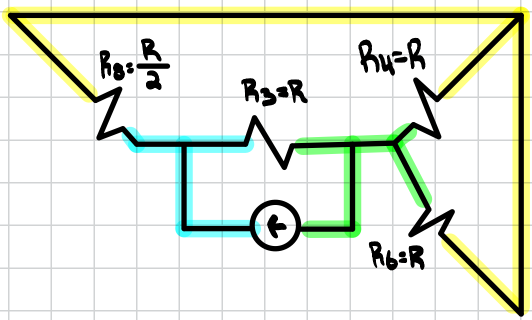
R_1 and R_7 will short because both ends are connected by the same node so you can remove them from the circuit



Cyan = Node A
Green = Node B
Yellow = Node C

R2 and R5 are in Parallel because both end of the resistors are connected by same node (Node A and C)

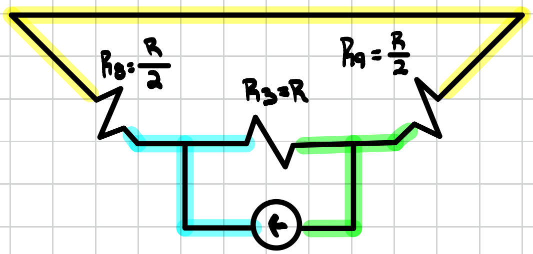
$$\left(\frac{1}{R}+\frac{1}{R}\right)^{-1}=\frac{R}{2}$$



Cyan = Node A
Green = Node B
Yellow = Node C

R4 and R6 are in Parallel for the same reason (Node B and C)

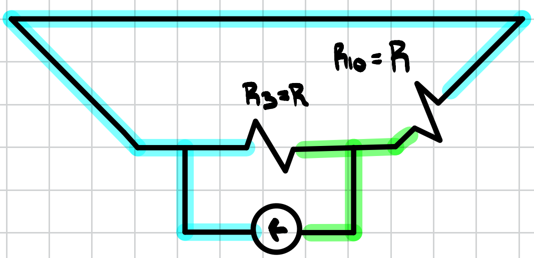
$$\left(\frac{1}{R}+\frac{1}{R}\right)^{-1}=\frac{R}{2}$$



Cyan = Node A
Green = Node B
Yellow = Node C

R_8 and R_9 are in series because the same current goes through it

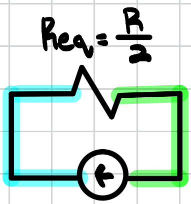
$$\frac{R}{2} + \frac{R}{2} = R$$



Cyan = Node A
Green = Node B

R_{10} and R_3 are in parallel since they share the same nodes (Node A and B)

$$\left(\frac{1}{R} + \frac{1}{R}\right)^{-1} = \frac{R}{2}$$



Cyan = Node A
Green = Node B